

FedALoRA: Adaptive Local LoRA Aggregation for Personalized Federated Learning in LLM

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Abstract—Federated large language model (FedLLM) shows excellent potential in collaboratively training large language models (LLMs) under the federated learning (FL) framework, which is benefiting from its privacy protection advantage. However, FedLLM faces the significant challenge of the nonidentical and independently distributed (non-IID) problem. In the real world, there are often cross-source or even cross-domain language set data between Internet of Things devices. To address the issue, we propose a new FedLLM framework FedALoRA via personalized and efficient parameter fine-tuning (PEFT). Specifically, the proposed scheme combines the personalized aggregation method and the LoRA method, which can adaptively aggregate the downloaded global model and local model to the local target on each client while ensuring low training costs. This adaptation initializes the local model before each iterative training, enabling clients to learn general knowledge while enhancing their understanding of their own domain knowledge. Extensive experiments and analysis on cross-domain non-IID settings and the financial datasets on Dirichlet non-IID settings demonstrate the effectiveness and superiority of FedALoRA.

Index Terms—Large language model (LLM), LoRA tuning, low communication, personalized federated learning (PFL).

I. INTRODUCTION

RECENTLY, large language models (LLMs), such as ChatGPT [1] and LLaMA, have attracted widespread attention in various domains. As LLMs become more and more common, it becomes increasingly necessary to customize them for domain-specific tasks. Take the scenario of federated instruction tuning as an example, the cooperation among

clients hailing from diverse domains is capable of augmenting the general competencies of LLMs, like the chatting proficiency. Meanwhile, each client also shows particular interest in its own domain, say, providing answers to financial questions. Such customization usually relies on supervised fine-tuning (SFT), in which the LLM is adapted to domain-specific data through various techniques [2]. The main current SFT techniques can be divided into three categories [3]: 1) fine-tuning; 2) parameter-efficient fine-tuning (PEFT); and 3) fast fine-tuning. LoRA is an effective method in PEFT technology. Compared with full fine-tuning, it has obvious advantages in reducing training parameters and computing costs. Together with fast fine-tuning, it constitutes a different fine-tuning technology path to serve LLM fine-tuning scenarios with different needs. LoRA has attracted much attention due to its effectiveness in PEFT. By optimizing the rank factorization matrix, LoRA can train the low-rank parameter matrix in the LLM while freezing the original parameters (see Section II).

The combination of federated learning (FL) and LLM provides a good opportunity for collaborative learning across clients [4], [5]. Tian et al. [6] combined split learning and FL to pretrain the BERT model. Similarly, Jiang et al. [7] demonstrated a low-parameter FL method based on low-rank adaptation for task fine-tuning of LLMs. Reference [8] addressed the problem of collaborative and off-site fine-tuning of LLMs through an open-source framework for comprehensive FL called “Fate-LLM.” However, the inherent challenge of data heterogeneity, i.e., data distribution across clients is nonidentical and independently distributed (non-IID), poses a significant obstacle [9], [10]. This frequently results in the subpar performance of the global model because of its convergence toward the client with the poorest performance, which is known as the “bucket effect” [11]. To address this challenge, recent efforts have combined personalized FL (PFL) with techniques, such as LoRA fine-tuning to train personalized local models for each client and alleviate the traditional “bucket effect” [3], [12], [13], [14]. PFL endeavors to customize the training process for each client, factoring in their distinctive data distribution patterns and learning objectives. In this way, the model becomes capable of seizing not only the cross-client wisdom, known as shared knowledge, but also the client-exclusive insights, termed personalized knowledge. Through the personalized training approach, PFL strives to alleviate the adverse consequences of data heterogeneity and bolster the performance of every single client. However, the huge computational and communication overheads involved in

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processing LLM parameters remain a significant challenge for these approaches.

In the realm of the Internet of Things (IoT), the need for intelligent data processing and model training has become increasingly crucial. IoT devices, such as smart sensors, wearable devices, and home appliances, generate vast amounts of data. These data are often non-IID, as different devices are deployed in diverse environments and serve different purposes. For example, in the financial domain [15], different intelligent financial terminal devices serve different user groups, and these users exhibit significant differences in aspects, such as investment strategies, risk preferences, levels of financial knowledge, and trading habits. Some users prefer long-term investments and focus on macroeconomic trends and the fundamentals of companies, while some users are keen on short-term speculation and pay more attention to market hotspots and technical indicators. These differences lead to variations in the characteristics and distributions of the financial-related data collected by the devices.

Applying the concept of FL to IoT can enable collaborative learning among these devices without sharing sensitive data directly. However, the existing challenges in FL for LLMs, such as the non-IID problem and high computational costs, are further exacerbated in the IoT context. IoT device deployment and data generation are inherently heterogeneous.

IoT devices are distributed in different environments (e.g., urban, rural, industrial environments) and serve different purposes (e.g., medical monitoring, smart transportation, home automation). Therefore, the data they collect naturally exhibit significant distribution differences, such as differences in feature space, label distribution, and temporal patterns. For example, smart sensors in factories may generate time series data with specific mechanical failure patterns, while smart sensors in hospitals may capture medical signals with specific patient anomalies. In addition, user behavior and preferences further amplify this non-IID property, taking wearable devices as an example, wearable devices worn by athletes and elderly users collect very different distributions of physiological data. This inherent diversity makes the non-IID problem more prominent in the IoT than in traditional FL settings.

IoT devices typically have limited computational resources, including low-power processors, small memory capacities, and restricted energy supplies. This makes it extremely challenging to implement complex model training algorithms. IoT networks typically operate with limited bandwidth, especially in rural or remote areas. LLMs generate large amounts of data during training, including model parameters and gradients. Transmitting these data between IoT devices and the server, or among devices in a peer-to-peer FL setup, can quickly exhaust the available bandwidth.

To address these challenges, we draw inspiration from previous literature [16] and integrate LLM custom processes into the personalized aggregation PFL framework. By doing so, we can extract insights from clients from different data distributions while maintaining privacy and improving performance. Each institution acts as a client and contributes its unique data to LLM training without directly sharing sensitive information, thus improving the stability of

their respective systems by effectively using distributed data sources.

In this article, we propose an LLM customization framework. By integrating LoRA fine-tuning and the personalized aggregation PFL framework (aka FedALoRA), it features adaptive local LoRA aggregation (ALoRA) and no additional communication. In the personalized aggregation PFL, there are N clients, each client fine-tunes the LLM locally using private data, and each client updates a copy of the LLM. After receiving the global model from the previous round, each client does not directly overwrite the local model with the global model, but selectively fuses the global model into the local model. Then the local training is performed based on this fused model. Specifically, the framework initializes an adaptive LoRA aggregation module for the LLM in each client, that is, a tradeoff is made between the local LoRA module and global LoRA module to capture the part of global knowledge that is more useful to the local client. The adaptive LoRA aggregation module of each client is independent, thus ensuring that the LLM adapts to the local data. After local training, the client uploads the parameters of the local LoRA module to the server for aggregation operations to integrate the knowledge of different clients. In addition, in order to retain the basic knowledge learned by the LLM from the large corpus, we freeze the parameters of the LLM. The overall framework of FedALoRA is shown in Fig. 1.

Our main contributions can be summarized as follows.

- 1) We propose FedALoRA, a novel adaptive LoRA aggregation and no-extra-communication framework, which has a fine-grained ALoRA module that aggregates global models and local models element by element to adapt to objectives in different domains. To the best of our knowledge, this is the first work to integrate LoRA and personalized aggregation PFL for LLM customization.
- 2) We conduct experiments in the cross-domain dataset setting and demonstrate excellent performance. Our approach can effectively improve performance in the cross-domain and finance domain, which keeps communication costs unchanged.
- 3) We explore the non-IID problem in federated LLMs (FedLLMs) when the client holds a cross-domain dataset and non-IID financial domain datasets, which can be seen as analogous to the diverse and nonhomogeneous data in IoT devices. These experiments not only show the potential of collaboration among diverse institutions but also provide a new solution for IoT-related model training.

The remaining portions of this article are organized as follows. Section II describes the related work of this article. Section III introduces the preliminaries of this article. Section IV describes the proposed scheme. Section V describes the performance analysis of proposed scheme. Conclusions can be found in Section VI.

II. RELATED WORKS

Early FL research often focused on data rebalancing techniques to alleviate the non-IID problem. For example, the

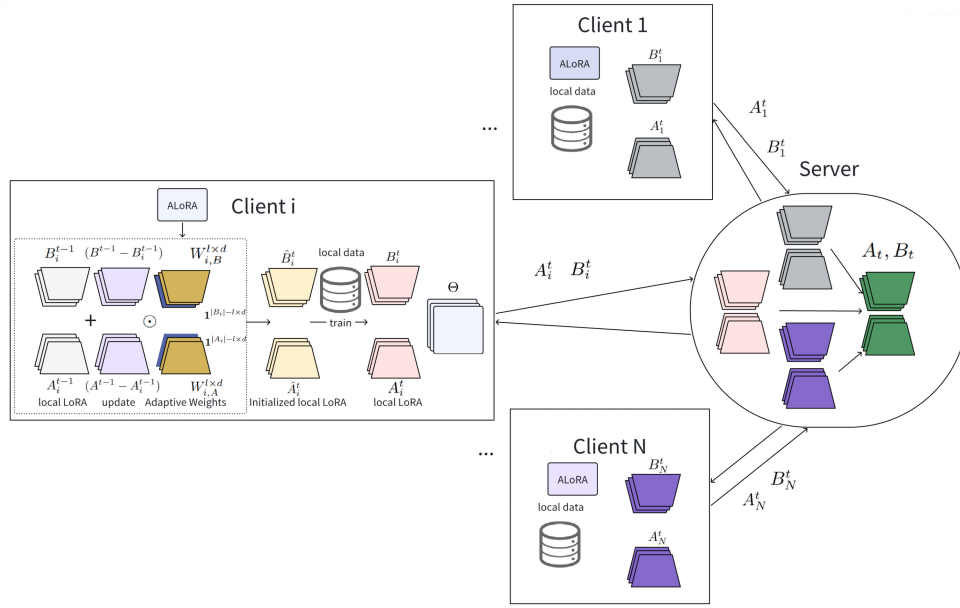


Fig. 1. Overview of FedALoRA framework.

basic FL algorithm FedAvg [4] uses a weighted average of model parameters to aggregate knowledge across clients. However, this approach has difficulties in dealing with severely non-IID data because it assumes that the data distributions of clients are similar enough. Subsequent studies, such as FedProx [17] introduced a proximal term to regularize local updates to the global model, thereby reducing client drift in heterogeneous scenarios, but sacrificed some personalization for global convergence.

To balance data diversity and model performance, PFL came into being. PFL aims to train client-specific models that retain both global shared knowledge and local domain expertise. Currently, pFL methods for model aggregation on the server side can be mainly divided into three categories: fine-tuning a single global model; learning additional personalized models; and learning local models through personalized (local) aggregation.

For Category 1, FedRep [18] splits the model backbone into a globally shared representation layer (global model) and a client-specific task head (local model). During training, the global model is aggregated and updated on the server side, and the client only fine-tunes the task head based on local data, thereby balancing global representation learning and client personalization needs.

For Category 2, pFedMe [19] learns an independent personalized model for each client based on Moreau Envelopes, and introduces a distance constraint with the global model when optimizing the local model. During training, the client achieves local model personalization while maintaining relevance to the global model by minimizing the objective function that includes the global model regularization term.

For Category 3, FedAMP [20] learns adaptive model parameters for each client by separating global and personalizable layers, allowing clients to customize higher layers based on their data while sharing common knowledge in lower

layers. FedALA [16] uses the adaptive local aggregation (ALA) module to adaptively aggregate the downloaded global model and the local model in an element-wise manner. This unique approach enables the aggregation result to be more aligned with local objectives, thereby providing more accurate information for the initialization process of local model training.

Fan et al. [8] proposed FATE-LLM, an open-source framework for industrial applications, whose objective is to address the efficiency challenges in federated IoT environments with different configurations. Under these configurations, techniques such as knowledge distillation, off-site tuning, various LLM architectures, and PEFT algorithms can be applied. In the testing experiment, researchers selected the ChatGLM-6B model and carried out fine-tuning operations on multiple clients based on the AdvertGen dataset using either the LoRA algorithm.

Ye et al. [21] proposed an integrated, concise and easy-to-research framework called OpenFedLLM. This framework covers federated instruction adjustment, federated value alignment, seven classic FL baseline algorithms, eight language training datasets, and more than 30 evaluation metrics. Based on OpenFedLLM, the authors conducted a comprehensive empirical analysis and demonstrated the advantages of participating in FL compared to local training alone. They also explored the direction of collaboration among institutions in multiple different domains and found that the level of FL algorithms in professional domains cannot beat expert clients, which made us think that this problem can be solved through PFL methods.

In the IoT scenarios, the non-IID situation of data is extremely common. At the same time, due to significant differences in functions, usage environments, and user requirements among various IoT devices, it is particularly necessary to tailor personalized strategies for each device. In addition, with the

development of artificial intelligence technology, LLMs are being increasingly applied in the field of the IoT. In view of the above background and the aforementioned work, we propose this scheme.

III. PRELIMINARIES

A. Federated Learning

To establish the FL framework, we employ FedAvg, a conventional SGD-based aggregation algorithm. Specifically, we assume that each client has its own local dataset, denoted as $D_i = (x, y)$, where x and y are the model's instruction and the expected response, respectively. The collective dataset across all clients is denoted as $D = \{D_1, \dots, D_N\}$, where N denoted the number of clients. At each communication round t , clients upload their local model parameters W_i^t to the server for aggregation

$$W^t = \sum_{i=1}^N \frac{|D_i|}{|D|} W_i^t \quad (1)$$

where W^t is the globally averaged parameter; $|D_i|$ is the client i sample size; and $|D|$ is the total sample size. The server then broadcasts the aggregated parameter W^t to each client for the next round of local training.

B. Personalized Federated Learning

In contrast to traditional FL, research into PFL centers around confronting two crucial challenges.

First, there is the thorny issue of convergence concerning heterogeneous data, which becomes especially conspicuous when engaging in learning tasks with Non-IID data. In the IoT, there are a large number of devices with various types, and the problem of data heterogeneity is further amplified. Different types of IoT devices have huge differences in the characteristics, distribution and scale of the data they collect, showing typical Non-IID characteristics. When conventional learning methodologies, exemplified by the ubiquitously adopted FedAvg algorithm, are utilized to train such data, the precision of the model witnesses a pronounced decline. The fundamental origin of this performance deterioration lies in the fact that the local training and synchronization procedures of non-IID data trigger the client drift phenomenon, impeding the model's progression toward the optimal global convergence.

Second, the dearth of personalized strategies poses a significant hurdle. The traditional IoT FL paradigm often focuses on training a general global model to fit the so-called "average device." However, in actual applications, each IoT device has its own unique functions and usage scenarios, and the local data distribution it faces may be very different from the global distribution. Nevertheless, when confronted with local distribution scenarios that diverge substantially from the global distribution, such shared models struggle to effectuate efficacious generalization. The deficiency of personalization becomes glaringly evident when handling the prevalent non-IID local datasets in practical applications, substantially impeding the model's generalization potential.

After all, in such intricate circumstances, a single all-purpose model generally falls short of fulfilling diverse demands.

To address the aforementioned challenges, PFL treats each device as a client in the IoT scenario and categorizes the knowledge of each client into two types: 1) general knowledge and 2) client-specific knowledge. In non-IID scenarios, the data distributions of different clients are unique. To utilize both types of knowledge simultaneously during the training process, recent methods [16], [22] try to generate client-specific models through personalized aggregation. Formally, this personalized aggregation can be formulated as

$$\hat{W}_i^t = P(W^{t-1}, W_i^{t-1}, h) \quad (2)$$

where $\hat{W}_i^t$ denotes the parameters to be trained by client i in the t round. $P(\cdot)$ denotes the function to personally aggregate global parameters and local parameters, and h denotes some hyperparameters to control aggregation. Consequently, the PFL that aims to accommodate statistical heterogeneity endeavors to achieve high precision for heterogeneous data of diverse types and degrees, all without prior knowledge of the data distribution.

C. Low-Rank Adaptation

IoT devices are usually resource-constrained, with limited computing power, storage capacity, and energy supply. Traditional model fine-tuning methods consume a lot of resources when dealing with LLMs with parameters ranging from millions to billions, and are difficult to run effectively on IoT devices. There is an urgent need to explore more cost-effective methods. Against this backdrop, researchers have recently innovatively proposed the PEFT method. The core objective of PEFT is to reduce the number of parameters involved in the fine-tuning process and lower the computational complexity on the one hand and to improve the performance of pretrained LLMs in downstream tasks and thus control the training costs on the other hand. In this way, even when computing resources are limited, LLMs can still flexibly adapt to downstream tasks, effectively achieving efficient transfer learning results and broadening the boundaries for the wide application of artificial intelligence.

Currently, LoRA stands out as the most prevalent PEFT method. Its edge over other techniques like adapter, prefix tuning, and traditional fine-tuning in terms of performance has propelled it to the forefront. Notably, LoRA has the distinct capability of embedding domain-specific knowledge into low-rank parameter matrices, all while leaving the original parameters intact.

Given a parameter matrix $W_0 \in \mathbb{R}^{d \times k}$, the low-rank decomposition can be utilized to represent parameter updates ΔW as follows:

$$W_0 + \Delta W = W_0 + BA \quad (3)$$

where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and $r \ll \min(d, k)$. W_0 is frozen in training stage, while A and B are trainable.

Through learning rank decomposition matrices and concurrently freezing the original parameters, LoRA achieves a significant reduction in the number of trainable parameters (as

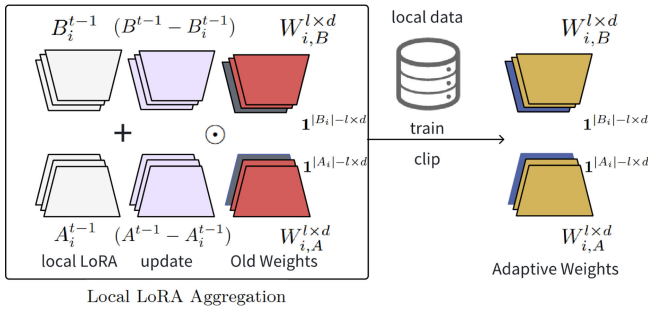
TABLE I

COMPARISON OF TRAINING PARAMETERS AND COMMUNICATION PARAMETERS BETWEEN TRADITIONAL FINE-TUNING, LoRA FINE-TUNING, AND OUR SCHEME

Method	tuning	LoRA	ALoRA
$N_{trainable}$	6738 M	4.194 M	4.194 M-8.388M
$N_{communication}$	6738 M	4.194 M	4.194 M

TABLE II
LIST OF KEY SYMBOLS

Notations	Description
N	Number of clients
D_i	Local dataset of client i
D	Global dataset ($D = \{D_1, \dots, D_N\}$)
W_i^t	Model parameters of client i at round t
$\bar{W}^t$	Aggregated global parameters at round t
Θ	Frozen pre-trained large language model parameters
A, B	Global LoRA parameter matrices ($A \in \mathbb{R}^{r \times k}, B \in \mathbb{R}^{d \times r}$)
A_i, B_i	Local LoRA parameters of client i
r	Rank parameter in LoRA
$\hat{A}_i^t, \hat{B}_i^t$	Adaptively aggregated LoRA parameters for client i at round t
$W_{i,A}, W_{i,B}$	Adaptive weight matrices for global-local LoRA fusion
l	Hyperparameter: number of top transformer layers for ALoRA
α	Dirichlet distribution parameter (controls non-IID severity)

Fig. 2. Learning process in ALoRA modules. A denotes “Adaptive.” Here, we consider a 32-layer model (such as Llama2-7b) and set $l = 20$.

shown in Table I. This leads to a substantial decrease in the storage needs of LLM variants optimized for specific tasks. Moreover, it streamlines the process of efficient task switching during deployment, without incurring any additional inference latency.

D. Notations

The notations and their semantic meanings utilized in this article to describe our scheme are presented in Table II.

IV. OUR SCHEME: FEDALORA

A. Problem Statement

The objective of this article is to train the LoRA module for a variety of clients by integrating the PFL technique. This approach aims to enhance performance while safeguarding data privacy and ensuring computational efficiency. Suppose we have N clients, each client has its own local dataset, denoted as $D_i = (x, y)$, where x and y are the model’s instruction and the expected response, respectively. The collective dataset across all clients is denoted as $D = \{D_1, \dots, D_N\}$, where each client has its private training data

D_i . These data are sampled from N distinct domains and have different sizes. With the help of a central server, our goal is to collaboratively learn individual local models, namely, $\Theta + \hat{B}_1^t \hat{A}_1^t, \dots, \Theta + \hat{B}_N^t \hat{A}_N^t$, for each client by making use of their respective datasets $D_1, \dots, D_N$, all while ensuring that the private data remains unexchanged. The ultimate goal is to minimize the global objective function and derive appropriate local models that are well-suited to every specific domain

$$\{\Theta + \hat{B}_1^t \hat{A}_1^t, \dots, \Theta + \hat{B}_N^t \hat{A}_N^t\} = \arg \min G(L_1, \dots, L_N) \quad (4)$$

where $L_i = L(\Theta + \hat{B}_i^t \hat{A}_i^t, D_i; \Theta + BA) \forall i \in [N]$ and $L(\cdot)$ is the loss function. Θ is the frozen pretrained model, B and A are the global LoRA parameters, which together form the global model. $\Theta + BA$ brings multiple domain information to client i . Typically, $G(\cdot)$ is set to $\sum_{i=1}^N k_i L_i$, where $k_i = |D_i|/|D|$ and $|D_i|$ is the number of local data samples on client i .

In the case where each client holds a fine-tuned dataset from different domains, the server generates a global model by aggregating the fine-tuned client models, but the professional level of this global model on each domain is not as good as the result of local training on the expert client, whom holds the dataset from this domain [21]. To solve this problem, we propose a scheme that combines LoRA technology and personalized aggregation technology, FedALoRA, which has a fine-grained ALoRA module that aggregates global models and local models element by element to adapt to objectives in different domains. We show the learning process in ALoRA in Fig. 2.

B. Framework Overview

We introduce the overview of the FedALoRA framework in this section. As shown in Fig. 1, each client is comprised of three main components: 1) LLM; 2) LoRA module; and 3) ALoRA module. LLM has the basic knowledge, the LoRA module is used to absorb knowledge and communication from distributed data sources, and the ALoRA module is used to make tradeoffs between the local LoRA module and the global LoRA module. Specifically, our scheme first trains the ALoRA module locally on clients. Then, the ALoRA module performs personalized aggregation between the previous local LoRA module and the global LoRA module to obtain an initialized local LoRA module. Clients then train based on this initialized local LoRA module to generate a new round of local LoRA modules. Next, clients upload their local LoRA modules to the server, which aggregates the uploaded LoRA modules from each client by weighting according to sample size to form a global LoRA module. Finally, the server distributes the global LoRA module to clients for the next training round.

C. ALoRA in FedALoRA

In traditional FedLLM (such as FedAvg+LoRA), after the servers send the old global LoRA module A^{t-1}, B^{t-1} to client i in iteration t , A^{t-1}, B^{t-1} overwrites the old local LoRA module A_i^{t-1}, B_i^{t-1} to obtain the initialized local LoRA module $\hat{A}_i^t, \hat{B}_i^t$ for local model fine-tuning, i.e., $\hat{A}_i^t := A^{t-1}, \hat{B}_i^t := B^{t-1}$. In FedALoRA, we element-wisely aggregate the global

LoRA module and local LoRA module rather than overwriting. Formally

$$\hat{A}_i^t = A^{t-1} \odot W_{i,A}^1 + A_i^{t-1} \odot W_{i,A}^2 \quad \text{s.t. } w_{1,A}^q + w_{2,A}^q = 1 \quad \forall \text{ valid } q \quad (5)$$

$$\hat{B}_i^t = B^{t-1} \odot W_{i,B}^1 + B_i^{t-1} \odot W_{i,B}^2 \quad \text{s.t. } w_{1,B}^s + w_{2,B}^s = 1 \quad \forall \text{ valid } s \quad (6)$$

where $\odot$ is a Hadamard product, and $w_{1,A}^q$ and $w_{2,A}^q$ are, respectively, the q th parameters in the aggregating weights $W_{i,A}^1$ and $W_{i,A}^2$. $w_{1,B}^s$ and $w_{2,B}^s$ are similar. When $\forall$ valid q , $w_{1,A}^q \equiv 1$, $w_{2,A}^q \equiv 0$, it is overwriting. However, it is hard to learn four weights matrix with the constraint through the gradient-based learning method. Thus, we combine $W_{i,A}^1$ and $W_{i,A}^2$ by viewing (7) and combine $W_{i,B}^1$ and $W_{i,B}^2$ by viewing (8) as

$$\hat{A}_i^t = A_i^{t-1} + (A^{t-1} - A_i^{t-1}) \odot W_{i,A} \quad (7)$$

$$\hat{B}_i^t = B_i^{t-1} + (B^{t-1} - B_i^{t-1}) \odot W_{i,B} \quad (8)$$

where we call the term $(A^{t-1} - A_i^{t-1})$ as the update of A . In order for $W_{i,A}$, $W_{i,B}$ to satisfy the previous constraint s , we utilize element-wise weight clipping $\sigma(w_A) = \min(1, \max(0, w_A))$, $\sigma(w_B) = \min(1, \max(0, w_B))$ for regularization [23] and let $w_A, w_B \in [0, 1] \quad \forall w_A \in W_{i,A} \quad \forall w_B \in W_{i,B}$.

Given that the lower layers within the DNN capture more generalized information compared to the higher layers, clients predominantly seek the information present in the lower layers of the global model [24]. To curtail computational expenditure, we incorporate a hyperparameter l . This parameter dictates the scope of ALoRA modules by implementing it on the LoRA module in l topmost transformer layers and overwriting the LoRA module in the lower transformer layers like FedAvg for local initialization

$$\hat{A}_i^t = A_i^{t-1} + (A^{t-1} - A_i^{t-1}) \odot [\mathbf{1}^{|A_i|-l \times d}; W_{i,A}^{l \times d}] \quad (9)$$

$$\hat{B}_i^t = B_i^{t-1} + (B^{t-1} - B_i^{t-1}) \odot [\mathbf{1}^{|B_i|-l \times d}; W_{i,B}^{l \times d}] \quad (10)$$

where $|A_i|$, $|B_i|$ are the number of LoRA modules in A_i^{t-1} , B_i^{t-1} , d is the number of LoRA modules used in each transformer layer, usually 2. $\mathbf{1}^{|A_i|-l \times d}$, $\mathbf{1}^{|B_i|-l \times d}$ have the same shape of the lower LoRA modules in A_i^{t-1} , B_i^{t-1} . The elements in $\mathbf{1}^{|A_i|-l \times d}$, $\mathbf{1}^{|B_i|-l \times d}$ are ones (constants). the weight $W_{i,A}^{l \times d}$, $W_{i,B}^{l \times d}$ have the same shape of the remaining higher LoRA modules in A_i^{t-1} , B_i^{t-1} .

We initialize the value of each element in $W_{i,A}$, $W_{i,B}$ to one in the beginning and learn $W_{i,A}$, $W_{i,B}$ based on old $W_{i,A}$, $W_{i,B}$ in each iteration. Client i trains $W_{i,A}$, $W_{i,B}$ through the gradient-based learning method

$$W_{i,A} \leftarrow W_{i,A} - \eta \nabla_{W_{i,A}} \mathcal{L}(\Theta + \hat{B}_i^t \hat{A}_i^t, D_i, \Theta + B^{t-1} A^{t-1}) \quad (11)$$

$$W_{i,B} \leftarrow W_{i,B} - \eta \nabla_{W_{i,B}} \mathcal{L}(\Theta + \hat{B}_i^t \hat{A}_i^t, D_i, \Theta + B^{t-1} A^{t-1}) \quad (12)$$

where η is the learning rate for weight learning. We freeze other trainable parameters in ALoRA, including the pretraining parameters and entire LoRA parameters (local and global). After local initialization, client i utilizes SFT to capture knowledge from local domain data.

By opting for a relatively small value of l , we are able to substantially cut down the quantity of trainable parameters in ALoRA while incurring only a minimal degradation in performance. The specifics are presented in Section IV. Generally speaking [25], A is initialized as a random matrix that satisfies the Gaussian distribution, and B is an all-zero matrix. Note that the gradient information of B is coupled with A and will be affected by the initial data of A . Similarly, the gradient information of A is coupled with B and cannot gradient descent in the first training because B is an all-zero matrix. The gradient descent equations for A and B are as follows:

$$A = A - \nabla_A \mathcal{L} = A - (B)^T \nabla_{\Theta} \mathcal{L} \quad (13)$$

$$B = B - \nabla_B \mathcal{L} = B - \nabla_{\Theta} \mathcal{L}(A)^T. \quad (14)$$

According to (13), we can see that if B is a matrix of all zeros, the gradient of A is also 0. The first round of gradient information often cannot reflect the characteristics of local domain data distribution well. Therefore, we choose to let the client do WRound round warm-up trainings by SFT before the first aggregation, so that A and B can better reflect the characteristics of local domain data distribution, so that through a convergent ALoRA modules training, we can get a stable and reusable $W_{i,A}^l$, $W_{i,B}^l$. Algorithm 1 presents the overall FL process in FedALoRA.

D. Analysis of ALoRA

For the sake of simplicity, we leave out $\sigma(\cdot)$ and set $l = |A_i|/d$ in this context, which does not interfere with the subsequent analysis. According to (9) and (11), we have

$$\begin{aligned} \nabla_{W_{i,A}} \mathcal{L}_i^t &= \nabla_{\hat{A}_i^t} \mathcal{L}_i^t \odot \nabla_{W_{i,A}} (\hat{A}_i^t) \\ &= (\hat{B}_i^t)^T \nabla_{\Theta + \hat{B}_i^t \hat{A}_i^t} \mathcal{L}_i^t \odot (A^{t-1} - A_i^{t-1}) \end{aligned} \quad (15)$$

where $\mathcal{L}_i^t$ represents $\mathcal{L}(\Theta + \hat{B}_i^t \hat{A}_i^t, D_i, \Theta + B^{t-1} A^{t-1})$, Substitute it into (11) to obtain

$$W_{i,A} \leftarrow W_{i,A} - \eta (A^{t-1} - A_i^{t-1}) \odot (\hat{B}_i^t)^T \nabla_{\Theta + \hat{B}_i^t \hat{A}_i^t} \mathcal{L}_i^t. \quad (16)$$

Substitute the updated $W_{i,A}$ back into (9) to obtain

$$\begin{aligned} \hat{A}_i^t &= A_i^{t-1} + (A^{t-1} - A_i^{t-1}) \odot (W_{i,A} - \eta (A^{t-1} - A_i^{t-1}) \\ &\quad \odot (\hat{B}_i^t)^T \nabla_{\Theta + \hat{B}_i^t \hat{A}_i^t} \mathcal{L}_i^t) \\ &= \hat{A}_i^t - \eta (A^{t-1} - A_i^{t-1}) \odot (A^{t-1} - A_i^{t-1}) \\ &\quad \odot (\hat{B}_i^t)^T \nabla_{\Theta + \hat{B}_i^t \hat{A}_i^t} \mathcal{L}_i^t. \end{aligned} \quad (17)$$

Similarly, according to (10) and (12), we have

$$\begin{aligned} \nabla_{W_{i,B}} \mathcal{L}_i^t &= \nabla_{\hat{B}_i^t} \mathcal{L}_i^t \odot \nabla_{W_{i,B}} (\hat{B}_i^t) \\ &= \nabla_{\Theta + \hat{B}_i^t \hat{A}_i^t} \mathcal{L}_i^t (\hat{A}_i^t)^T \odot (B^{t-1} - B_i^{t-1}) \end{aligned} \quad (18)$$

$$\begin{aligned} \hat{B}_i^t &= B_i^{t-1} + \eta (B^{t-1} - B_i^{t-1}) \odot (B^{t-1} - B_i^{t-1}) \\ &\quad \odot \nabla_{\Theta + \hat{B}_i^t \hat{A}_i^t} \mathcal{L}_i^t (\hat{A}_i^t)^T. \end{aligned} \quad (19)$$

Algorithm 1: FedALoRA

Input : N clients, D : dataset across all clients,
 $WRound$: client warming round, L : loss
function, Θ : freezed pretrained model, A_0, B_0 :
initial global LoRA params, η : the learning
rate in ALoRA, and $\sigma(\cdot)$: clip function.

Output: Reasonable local
models: $\Theta + \hat{B}_1^t \hat{A}_1^t, \dots, \Theta + \hat{B}_N^t \hat{A}_N^t$

Server sends Θ, A_0, B_0 to all clients to initialize local
models

Clients initialize $W_{i,A}^l, W_{i,B}^l, \forall i \in [N]$ to ones

for i to N **do**

for i to $WRound$ **do**

$D_i^{(s)} \leftarrow$ Randomly sample s data from D_i

$A_i, B_i \leftarrow SFT(f(D_i^{(s)}; \Theta, A_i, B_i))$

end

end

Server obtains A, B

by $A \leftarrow \sum_i k_i A_i, B \leftarrow \sum_i k_i B_i$, where $k_i = |D_i|/|D|$

for iteration $t = 1, \dots, T$ **do**

Server sends Θ, A, B to all clients.

for i to N **do**

$D_i^{(s)} \leftarrow$ Randomly sample s data from D_i

if $t = 1$ **then**

while $W_{i,A}^l, W_{i,B}^l$ does not converge **do**

client i trains $W_{i,A}^l, W_{i,B}^l$ by (11) and
(12), there $D_i \leftarrow D_i^{(s)}$

client i clips $W_{i,A}^l, W_{i,B}^l$ by using $\sigma(\cdot)$

end

end

else if $t > 1$ **then**

client i trains $W_{i,A}^l, W_{i,B}^l$ by (11) and (12),
there $D_i \leftarrow D_i^{(s)}$

client i clips $W_{i,A}^l, W_{i,B}^l$ by using $\sigma(\cdot)$

end

Client i obtains $\hat{A}_i^t, \hat{B}_i^t$ by (9) and (10)

Client i obtains A_i^t, B_i^t by

$A_i^t, B_i^t \leftarrow SFT(f(D_i^{(s)}; \Theta, \hat{A}_i^t, \hat{B}_i^t))$

Client i sends A_i^t, B_i^t to the server.

end

Server obtains A, B by

$A \leftarrow \sum_i k_i A_i^t, B \leftarrow \sum_i k_i B_i^t$, where $k_i = |D_i|/|D|$

end

return $\Theta + \hat{B}_1^t \hat{A}_1^t, \dots, \Theta + \hat{B}_N^t \hat{A}_N^t$

So we can view updating $W_{i,A}, W_{i,B}$ as updating $\hat{A}_i^t, \hat{B}_i^t$ in ALoRA. In iteration t , the gradient $\nabla_{\Theta + \hat{B}_i^t \hat{A}_i^t} \mathcal{L}_i^t$ is scaled element-wise. In contrast to local model fine-tuning that only focuses on the local data of a single domain, the whole process of updating in (17) and (19) can be aware of the generic information in the global LoRA modules, which absorbed knowledge from multiple domains. Although the term $(A^{t-1} - A_i^{t-1}), (B^{t-1} - B_i^{t-1})$ both are constants in iteration, it can provide dynamic information for the ALoRA modules Learning to help it adapt to complex scenarios.

V. PREFORMATION ANALYSIS

In this section, we explore the non-IID problem in federated large language models when the client holds a cross-domain dataset and non-IID financial domain datasets, which can be seen as analogous to the diverse and nonhomogeneous data in IoT. These experiments in the cross-domain setting verify the effectiveness of the FedALoRA approach in collaboration between different institutions that have private datasets from different domains. The experiments in the non-IID setting on finance domain also demonstrate the effectiveness of FedALoRA under heterogeneous client datasets.

A. Experimental Setup

1) *Dataset*: To evaluate our approach and baselines on different IOT scenarios. We selected two scenarios: multidomain institutional collaboration and financial sentiment analysis. Specifically, we explore four exemplary datasets from Hugging Face, covering diverse domains (i.e., general, code, math and finance). Table III presents the descriptions of these datasets, encompassing information such as the domain, applied scenario, the number of samples, input length distribution, the average length of instruction, and the average length of response.

2) *Characteristics*: In Scenario – 1, the sample sizes of the four datasets show significant differences, with the sample sizes of each dataset ranging from 20k to 225k. The distribution of input lengths within these datasets varies greatly, which fully demonstrates the rich diversity of data features and also implies the potential challenges that may be encountered when dealing with inputs of different lengths. In Scenario – 2, the financial dataset used also exhibits a diverse distribution of input lengths, and this phenomenon further reflects the prevalent heterogeneity problem in the FL environment. The analysis results indicate that, in order to effectively ensure that the model has robust performance in the FL scenario, it is imperative to effectively handle the complex and diverse data features within it.

3) *Data Preprocessing*: Since the structure of each dataset is different, in order to unify the structure, we use the Alpaca template to format the datasets, which are mapped into instruction-response pairs. In addition, in Scenario–2, we employed the Dirichlet non-IID setting to simulate the cross-source scenarios of IoT devices. This setting is a widely adopted framework in current FL research, as shown in Reference [27]. In this framework, the data of each participating client is generated according to the Dirichlet distribution, denoted as $\text{Dir}(\alpha)$. Among them, the parameter α determines the shape of the distribution. When α increases, the degree of class imbalance in each client's dataset is alleviated accordingly. Therefore, the Dirichlet non-IID setting helps us evaluate the performance of the proposed scheme in various non-IID scenarios, thereby forming a deeper understanding of the robustness and adaptability of the scheme.

4) *Evaluation Metrics*: To ensure the effectiveness of FedALoRA, we adopt different metrics in two scenarios. In Scenario – 1, we use MMLU [28] for general evaluation, GSM8K [29] for math evaluation, HumanEval for code evaluation, and FPB [30] for finance evaluation. In Scenario – 2,

TABLE III
DATASET DESCRIPTIONS IN THIS ARTICLE: WE PRESENT THE DATASET'S DATA SAMPLE COUNT (N_{SAMPLE}), INPUT LENGTH DISTRIBUTION, AVERAGED INSTRUCTION LENGTH ($\bar{L}_{\text{INST}}$), AND AVERAGED RESPONSE LENGTH ($\bar{L}_{\text{RESP}}$)

Dataset Name	Domain	Applied Scenario	N_{sample}	Input length distribution	$\bar{L}_{\text{inst}}$	$\bar{L}_{\text{resp}}$
Alpaca	General	Scenario-1	52k	9-489	21	66
FinGPT [26]	Finance	Scenario-1 and Scenario-2	77k	2-2.11k	61	3
Code-Alpaca	Code	Scenario-1	20k	12-1.42k	69	100
MathInstruct	Math	Scenario-1	225k	16-5.07k	85	181

TABLE IV
PERFORMANCE COMPARISON BETWEEN FL METHODS ON DIFFERENT DOMAINS

Domain Evaluation	General		Math		Code		Finance	
	Accuracy	gain	Accuracy	gain	Accuracy	gain	Accuracy	gain
local	43.56	-	15.39	-	16.46	-	69.55	-
FedAvg	45.05	1.49	15.16	-0.23	15.24	-1.22	63.70	-5.85
FedProx	45.36	1.8	15.31	-0.08	14.02	-2.44	67.41	-2.41
SCAFFOLD	45.42	1.86	14.40	-0.99	15.24	-1.22	63.04	-6.51
FedALoRA($l = 20$)	45.58	2.02	16.30	0.91	17.07	0.61	78.13	8.58

the local data of each client is divided into a training set and a test set at a ratio of 9:1. The test set is stored locally on each client and follows the same distribution as the local training set.

5) *Baselines*: To evaluate the performance of FedALoRA, we compare it with four baselines. The details of each algorithm are as follows.

- 1) *Local*: The local LLM of each client conducts independent training using its local training set.
- 2) *FedAvg* [4]: It is a well-known FL algorithm that mainly aggregates model parameters through weighted averaging. Its core operation process is that each local model uploads its parameters to the server. The server calculates the mean value of all the uploaded model parameters and then sends this mean value back to all the local devices. This process can be iterated multiple times until the model converges.
- 3) *FedProx* [17]: An algorithm aimed at addressing the data heterogeneity issue in FL. Compared with FedAvg, a proximal term is introduced during local training. While minimizing the local loss function, a penalty term related to the deviation from the global model parameters is added, guiding the local model to approach the global model and improving the aggregation effect and convergence speed.
- 4) *SCAFFOLD* [31]: An algorithm used to solve the problem of inconsistent training caused by data heterogeneity among clients and different local update step sizes. During the training process, a personalized control variate is introduced for each client. During local training, the update direction is adjusted by combining with the control variate. When aggregating the models, the server integrates the update information and the corresponding control variate information, promoting the efficient and stable convergence of the global model in a heterogeneous environment.

All baselines here adopt the parameter-efficient fine-tuning method of LoRA. This is because traditional fine-tuning will bring too much computing pressure to IoT devices, and too much communication pressure to devices in FL scenarios. (see Table I).

6) *Training Details*: We use the pretrained Llama2-7B [32] as the base model, which is quantized by int8 for computation efficiency. Each client is trained for ten steps using the AdamW optimizer in each round. The maximum sequence length is set to 512. The experiments are conducted on an NVIDIA GeForce RTX 3090. The initial learning rate in the first round is $5e-5$, and in the last round it dropped to $1e-6$. The batch size is set to 16. The rank of LoRA [25] is 32 with a scalar $\alpha = 64$.

In Scenario-1, we have four clients, each of which saves 5000 data samples from four different datasets, and each baseline runs 100 rounds of FL. For FedALoRA, we set W Round to 10, which means 90 rounds of federated training after ten rounds of local training.

In Scenario-2, we have four clients, following the Dirichlet non-IID setting, and each client saves 5000 data samples from the FinGPT dataset. Here we classify the data into four categories based on label classification: positive, neutral, negative, and others, with parameter $\alpha \in \{0.1, 0.5, 1\}$. Each baseline runs 50 rounds of FL methods. For FedALoRA, we set W Round to 5, which means 45 rounds of federated training after five rounds of local training.

B. Experimental Results

1) *Performance Comparison*: Test models: The local method tested four local models, the FedALoRA scheme tested four personalized models, and the other FL methods tested global models. In scenario-1, Table IV reports the performance comparison of four locally trained models (testing only their specific domains) with the model trained using the baseline method in 4 domains. Table V reports the performance comparison of four models trained by four clients individually and four personalized models trained by FedALoRA. In scenario-2, Table VI reports the performance comparison of four locally trained models with the model trained using the baseline method under different Dirichlet parameter α settings for financial datasets. From the tables, we can draw the following conclusions.

- 1) *FedALoRA*: In scenario-1, our proposed scheme FedALoRA beat all the compared methods in all

TABLE V
PERFORMANCE COMPARISON BETWEEN FEDALoRA($l = 20$) AND LOCAL ON DIFFERENT DOMAINS. THE FOUR CLIENTS ARE TRAINED ON GENERAL, MATH, CODE, AND FINANCE DATASETS, RESPECTIVELY

Domain	General		Math		Code		Finance	
Evaluation	Accuracy	gain	Accuracy	gain	Accuracy	gain	Accuracy	gain
Client1(local)	43.56	-	15.24	-	14.63	-	36.55	-
Client1(FedALoRA)	45.58	2.22	15.54	0.30	14.02	-0.61	42.80	6.25
Client2(local)	44.67	-	15.39	-	14.02	-	43.07	-
Client2(FedALoRA)	45.22	0.55	16.30	0.91	14.02	0	44.20	1.13
Client3(local)	44.21	-	14.78	-	16.46	-	56.60	-
Client3(FedALoRA)	45.39	1.18	15.09	0.31	17.07	0.61	58.50	1.90
Client4(local)	45.46	-	15.31	-	11.59	-	69.55	-
Client4(FedALoRA)	45.68	0.22	15.69	0.38	14.02	2.43	78.13	8.58

domains and achieved the best performance in terms of accuracy, which significantly demonstrates the effectiveness of FedALoRA in personalization strategy. In the general domain test, our scheme performs better than other FL methods because each client can learn more useful knowledge from other domains, thereby focusing on the local domain, thereby improving the parameter quality in the LoRA module and providing more useful information to other clients, forming a virtuous circle. In the professional domain test, other FL methods are inferior to expert clients, and only FedALoRA performs better than expert clients in specific domains, and its superiority in the financial domain is particularly obvious. This shows that financial domain clients can learn more knowledge from other domains. From the Table V, we can observe that in the general test, the personalized models were effectively improved, which proves that multiinstitutional collaboration can improve the general capabilities of the model. Overall, personalized models outperform local models in all specific domains, which shows the effectiveness of cooperation between different institutions or synergy between diverse domains IOT devices to comprehensively improve the every professional performance of models. In scenario-2, FedALoRA beats all the comparison methods under all parameter settings, and the average accuracy among 4 clients significantly exceeds other FL methods and beats local methods, which significantly proves that FedALoRA can adapt to different degrees of non-IID settings and adapt well to heterogeneous data distributions. When the data heterogeneity is low ($\alpha \in \{0.1, 0.5\}$), our scheme can generate a model with the best performance for each client's unique data distribution, that is, it can fully absorb local knowledge while effectively utilizing global knowledge. When the data heterogeneity is high ($\alpha = 1$), our scheme can still generate a model that is better than local training on most client devices, which is something other FL methods almost impossible to do.

- 2) *FedAvg*: In scenario-1, FedAvg outperforms the local client in general tests, reflecting the effectiveness of collaboration, but it is inferior to the expert client in professional domain tests, indicating that the lack of traditional FL personalization strategy has an impact. In scenario-2, FedAvg is the worst baseline method,

with the lowest average accuracy under all parameter settings. Especially in the case of highly heterogeneous data distribution ($\alpha = 1$), FedAvg's performance has dropped drastically, which fully demonstrates the defects caused by the lack of traditional FL in dealing with data heterogeneity.

- 3) *FedProx*: In scenario-1, FedProx performs best overall among all FL methods, and has a small gap with expert clients in professional domain tests, especially in the financial domain. We speculate that this is because the proximal term introduced by FedProx effectively alleviates the negative impact of nonindependent and identically distributed data in different domains. In scenario-2, FedProx performs steadily on various clients, and the average accuracy under various parameter settings exceeds FedAvg, and can beat local clients on some clients. The proximal term still plays a significant role.
- 4) *SCAFFOLD*: In scenario-1, SCAFFOLD performs worse than FedAvg in the professional domain, indicating that its local control variables cannot play a good role. In scenario-2, the situation changes, and SCAFFOLD becomes the FL baseline method with the best overall performance, especially in the high heterogeneity data distribution ($\alpha = 1$), the generated global model can be comparable to or even exceed most local clients, becoming a universal global model.

In general, our scheme demonstrates excellent performance advantages and effective personalization strategies in both scenario 1 and scenario 2, which fully proves that FedALoRA can effectively alleviate data heterogeneity, improve model performance, and provide a unique personalization strategy for each device, so that the model can better adapt to its local data distribution.

2) *Computation Overhead*: We recorded the total time cost of each method in two scenarios, as shown in Table VII. In scenario-1, except for the 19-min cost in the initial stage, the cost of FedALoRA in each iteration is 3.69 min. In other words, ALoRA only takes an additional 1.26 min to achieve a significant improvement in accuracy. In scenario-2, except for the 22-min cost in the initial stage, FedALoRA only takes an additional 0.37 min in each iteration. Overall, the cost per iteration is similar to that of FedProx, which needs to calculate the proximal term, and even has a certain cost advantage in scenario-1.

TABLE VI
PERFORMANCE COMPARISON BETWEEN FL METHODS ON FINANCE DOMAIN ON THE DIRICHLET NON-IID SETTING, HERE $l = 20$

α	0.1					0.5					1				
client	client1	client2	client3	client4	Avg	client1	client2	client3	client4	Avg	client1	client2	client3	client4	Avg
local	99.60	96.80	99.00	79.80	93.80	93.80	91.80	84.60	88.80	89.75	80.60	65.40	70.80	84.60	75.35
FedAvg	73.80	96.00	99.00	75.60	86.10	86.60	87.80	83.60	83.40	85.35	70.60	59.20	52.40	23.40	51.40
FedProx	83.80	95.00	98.00	75.80	88.15	88.60	90.80	83.80	88.60	87.95	79.80	66.40	69.60	71.00	71.70
SCAFFOLD	96.00	95.00	98.00	77.80	91.70	80.40	89.80	83.60	87.60	85.35	80.60	67.00	71.20	69.80	72.15
FedALoRA	99.80	98.00	99.80	82.80	95.10	96.00	94.00	84.80	89.80	91.15	81.20	68.20	71.40	84.00	76.20

TABLE VII
COMPUTATION OVERHEAD ON TWO SCENARIOS AND THE COMMUNICATION OVERHEAD (TRANSMITTED PARAMETERS PER ITERATION).
 $\sum$ IS THE PARAMETER AMOUNT IN THE LoRA MODULES

	Computation				Communication
	Scenario – 1		Scenario – 2		Scenario – 1 and Scenario – 2 Param./iter.
	Total time	Time/iter.	Total time	Time/iter.	
local	239 min	2.39 min	95 min	1.9 min	0
FedAvg	243 min	2.43 min	94 min	1.88 min	$2 * \sum$
FedProx	401 min	4.01 min	112 min	2.24 min	$2 * \sum$
SCAFFOLD	242 min	2.42 min	95 min	1.9 min	$4 * \sum$
FedALoRA(l = 20)	24 + 19 + 332 min	3.69 min	10 + 22 + 102 min	2.27 min	$2 * \sum$

3) *Communication Overhead*: We show the communication overhead of a client in one round in Table VII. The communication overhead of our scheme is the same as that of FedAvg, which only uploads and downloads a model. SCAFFOLD needs to transmit personalized control variation information in addition to the model parameters. The dimension of the control variation is usually the same as the model parameter vector dimension, which doubles the amount of data for each communication.

4) *Convergence Analysis of FedALoRA*: FedALoRA's adaptive LoRA aggregation modules aim to aggregate the element-wise local and global LoRA modules, which is different from the traditional FedAvg method that directly overwrites. To understand the impact of this difference on model performance, we compare FedALoRA with the standard FL aggregation scheme and evaluate its performance.

For Scenario-1, Fig. 3 shows the training loss curves of other FL methods and FedAvg with the ALoRA module. On clients 1 and 3, the loss curves of other FL methods and FedAvg with the ALoRA module are very close, indicating that their performance is comparable. The loss curves of FedAvg with the ALoRA module on clients 2 and 4 are significantly lower than that of other FL methods, indicating that our scheme converges better. In short, we experimentally demonstrate that FedALoRA's adaptive LoRA aggregation modules do not negatively affect model convergence compared to traditional FL methods.

For Scenario-2, from Fig. 4, we can see that as the number of communication rounds increases, the accuracy of each method will increase. Our scheme is basically better than other baselines in each communication round. At the same time, the accuracy of each method tends to converge around 40–50 rounds, and our scheme also shows the same convergence.

5) *Effects of Hyperparameter l* : In Scenario-1, we tested the case of $l = 3, 12, 20, 32$ separately. By reducing the hyperparameter l , we can reduce the range of ALoRA, as shown in Table VIII. When l is reduced from 32 to 3, the

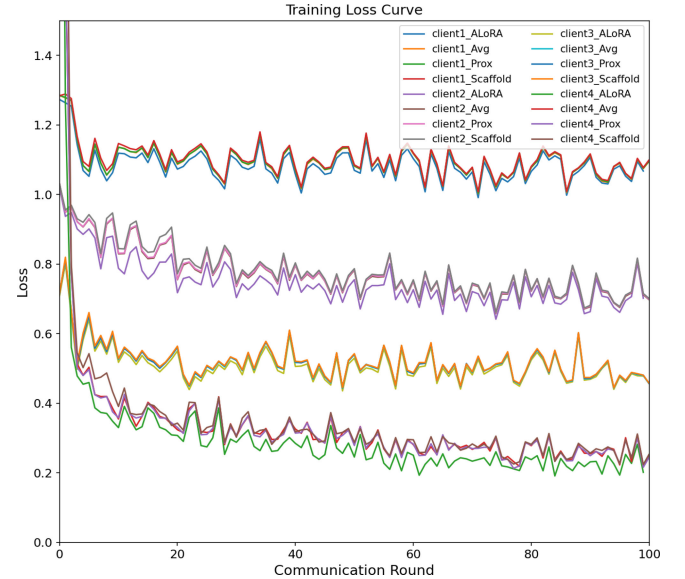


Fig. 3. Training loss of all FL methods in four clients across 100 communication rounds in Scenario-1.

TABLE VIII
PERFORMANCE COMPARISON OF FEDALORA UNDER FOUR DOMAINS VARYING l IN SCENARIO-1

Evaluation	trainable param.	General	Math	Code	Finance
$l=3$	0.393 M	45.30	15.24	15.24	71.80
$l=12$	1.573 M	45.68	15.54	15.85	72.60
$l=20$	2.621 M	45.58	16.30	17.07	78.13
$l=32$	4.194 M	45.44	16.45	17.07	77.14

number of trainable parameters in ALoRA will also decrease, because the number of parameters in each LoRA module is the same, and setting it to l is equivalent to training all LoRA parameters of l/l_{sum} (here $l_{\text{sum}} = 32$). We observed that the accuracy of each test will reach a maximum value around $l = 20$ and decrease to a certain extent when $l = 3$, which

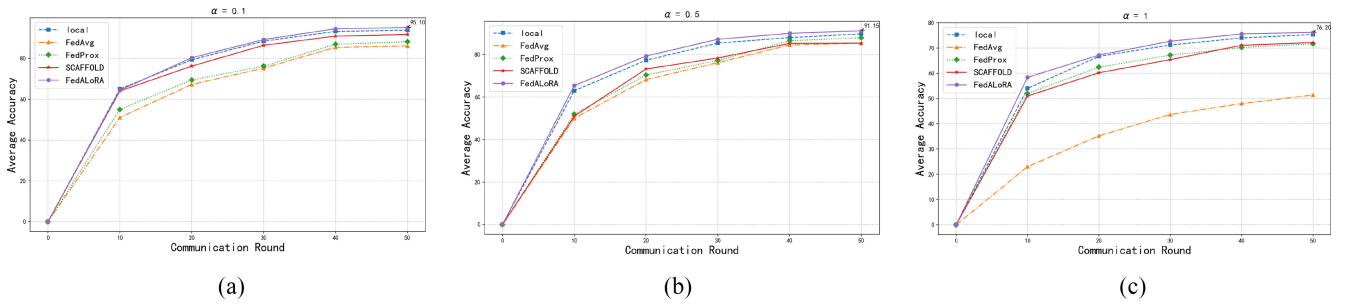


Fig. 4. Average Accuracy Curve between FL methods on finance domain on the Dirichlet non-IID setting, here $l = 20$. (a) $\alpha = 0.1$. (b) $\alpha = 0.5$. (c) $\alpha = 1$.

indicates that too little l will lead to less general knowledge learned.

VI. CONCLUSION

In this article, we propose FedALoRA, a novel ALoRA and no-extra-communication framework, which can capture more useful information in the global LoRA module. Our core idea is to enable clients to coordinate the large model fine-tuning process based on the ALoRA module. In FedALoRA, each client can effectively capture common knowledge across multiple domains as additional information to promote personalized learning and protect local large model. Our work not only provides a new direction for collaborative training scenarios of multidomain institutions, but also explores the possibilities of diverse IoT device scenarios and financial terminal device scenarios.

REFERENCES

- [1] T. Wu et al., "A brief overview of ChatGPT: The history, status quo and potential future development," *IEEE/CAA J. Automatica Sinica*, vol. 10, no. 5, pp. 1122–1136, May 2023.
- [2] F. Leon, M. Hulea, M. Gavrilescu, G. Kim, J. Yoo, and S. Kang, "Efficient federated learning with pre-trained large language model using several adapter mechanisms," *Mathematics*, vol. 11, no. 21, p. 4479, 2023.
- [3] X. Wu, X. Liu, J. Niu, H. Wang, S. Tang, and G. Zhu, "FedLoRA: When personalized federated learning meets low-rank adaptation," in *Proc. ICLR*, 2024, pp. 1–14. [Online]. Available: <https://openreview.net/forum?id=bZh06ptG9r>
- [4] H. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Arcas, "Communication-efficient learning of deep networks from Decentralized data," 2016, *arXiv:1602.05629*.
- [5] J. Zhang et al., "Towards building the federated GPT: Federated instruction tuning," 2023, *arXiv:2305.05644*.
- [6] Y. Tian et al., "FedBERT: When federated learning meets pre-training," *ACM Trans. Intell. Syst. Technol.*, vol. 13, no. 4, pp. 1–26, 2022.
- [7] J. Jiang, X. Liu, and C. Fan, "Low-parameter federated learning with large language models," in *Proc. 21st Int. Conf. WISA*, Jul. 2023, pp. 319–330.
- [8] T. Fan et al., "FATE-LLM: A industrial grade federated learning framework for large language models," 2023, *arXiv:2310.10049*.
- [9] M. Chen, M. Jiang, Q. Dou, Z. Wang, and X. Li, "FedSoup: Improving Generalization and Personalization in federated learning via selective model interpolation," in *Proc. Int. Conf. Med. Image Comput. Comput. Assist. Interv.*, Jul. 2023, pp. 318–328.
- [10] X. Ma, J. Zhang, S. Guo, and W. Xu, "Layer-wised model aggregation for personalized federated learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun 2022, pp. 10082–10091.
- [11] J. Bai, D. Chen, B. Qian, L. Yao, and Y. Li, "Federated fine-tuning of large language models under heterogeneous language tasks and client resources," in *Proc. 38th Conf. Neural Inf. Process. Syst.*, Feb. 2024, pp. 1–27.
- [12] L. Yi, H. Yu, G. Wang, and X. Liu, "pFedLoRA: Model-heterogeneous Personalized federated learning with LoRA tuning," 2023, *arXiv:2310.13283*.
- [13] J. Qi, Z. Luan, S. Huang, C. Fung, H. Yang, and D. Qian, "FDLoRA: Personalized federated learning of large language model via dual LoRA tuning," 2024, *arXiv:2406.07925*.
- [14] F. Jiang et al., "Personalized wireless federated learning for large language models," 2025, *arXiv:2404.13238*.
- [15] J.-Y. Zheng, H. Zhang, L. Wang, W. Qiu, H.-W. Zheng, and Z.-M. Zheng, "Safely learning with private data: A federated learning framework for large language model," in *Proc. Conf. Empir. Methods Natural Lang. Process.*, Miami, FL, USA, Nov. 2024, pp. 5293–5306. [Online]. Available: <https://aclanthology.org/2024.emnlp-main.303/>
- [16] J. Zhang et al., "FedALA: Adaptive local aggregation for personalized federated learning," in *Proc. AAAI Conf. Artif. Intell.*, Dec. 2022, pp. 11237–11244.
- [17] L. Tian, A. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," 2018, *arXiv:1812.06127*.
- [18] L. Collins, H. Hassani, A. Mokhtari, and S. Shakkottai, "Exploiting shared representations for Personalized federated learning," 2021, *arXiv:2102.07078*.
- [19] C. T. Dinh, N. H. Tran, and T. D. Nguyen, "Personalized federated learning with Moreau envelopes," 2020, *arXiv:2006.08848*.
- [20] Y. Huang et al., "Personalized federated learning: An attentive collaboration approach," 2020, *arXiv:2007.03797*.
- [21] R. Ye et al., "OpenFedLLM: Training large language models on decentralized private data via federated learning," in *Proc. 30th ACM SIGKDD Conf. Knowl. Discovery Data Min.*, 2024, pp. 6137–6147.
- [22] X.-C. Li, D.-C. Zhan, Y. Shao, B. Li, and S. Song, "FedPHP: Federated personalization with inherited private models," in *Proc. Joint Eur. Conf. Mach. Learn. Knowl. Discovery Databases*, Jan. 2021, pp. 587–602. [Online]. Available: http://dx.doi.org/10.1007/978-3-030-86486-6_36
- [23] M. Arjovsky, S. Chintala, and L. Bottou, "Wasserstein generative adversarial networks," in *Proc. 34th Int. Conf. Mach. Learn.*, 2017, pp. 214–223.
- [24] J. Yosinski, J. Clune, Y. Bengio, and H. Lipson, "How transferable are features in deep neural networks," 2024, *arXiv:1411.1792*.
- [25] E. J. Hu et al., "LoRA: Low-rank adaptation of large language models," 2021, *arXiv:2106.09685*.
- [26] B. Zhang, H. Yang, and X.-Y. Liu, "Instruct-FinGPT: Financial sentiment analysis by instruction tuning of general-purpose large language models," 2023, *arXiv:2306.12659*.
- [27] T. Lin, L. Kong, S. Stich, and M. Jaggi, "Ensemble distillation for robust model fusion in federated learning," in *Proc. 34th Conf. Neural Inf. Process. Syst.*, Jan. 2020, pp. 1–13.
- [28] D. Hendrycks et al., "Measuring massive multitask language understanding," 2020, *arXiv:2009.03300*.
- [29] K. Cobbe et al., "Training verifiers to solve math word problems," 2021, *arXiv:2110.14168*.
- [30] P. Malo, A. Sinha, P. Takala, P. Korhonen, and J. Wallenius, "Good debt or bad debt: Detecting semantic orientations in economic texts," *J. Assoc. Inf. Sci. Technol.*, vol. 65, no. 4, pp. 782–796, Jul. 2013.
- [31] S. Karimireddy, S. Kale, M. Mohri, S. Reddi, S. Stich, and A. Suresh, "SCAFFOLD: Stochastic controlled averaging for federated learning," 2019, *arXiv:1910.06378*.
- [32] H. Touvron et al., "Llama 2: Open foundation and fine-tuned chat models," 2023, *arXiv:2307.09288*.



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